

SAMPLE RUN - A1

PART 1

```
wissemoumsalem@Wissems-MacBook-Air comp371-A1 % g++ -std=c++11 part1.cpp -o part1
./part1
```

```
=====
                COMP 371 - COMPUTER GRAPHICS
                ASSIGNMENT 1
                PART 1
=====
```

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Dynamic Array Creation and Memory Management Program

=====

ARRAY CREATION

=====

Enter the number of elements for the array: abc

Invalid input.

Please enter a positive whole number.

Enter the number of elements for the array: 5

Creating a dynamic array containing 5 elements...
Array successfully created.

Initializing the array with values from 0 to 4...
Array successfully initialized.

Here is the array:

[0, 1, 2, 3, 4]

The array is no longer needed.

Deleting the array and releasing its memory...

Memory successfully deleted and released.

=====

Would you like to create another array?

[Y] Yes, create another array

[N] No, end the program

Enter your choice: y

Starting a new array operation...

=====

ARRAY CREATION

=====

Enter the number of elements for the array: -2

Invalid input.

Please enter a positive whole number.

Enter the number of elements for the array: 3

```
Creating a dynamic array containing 3 elements...
Array successfully created.
```

```
Initializing the array with values from 0 to 2...
Array successfully initialized.
```

```
Here is the array:
```

```
[0, 1, 2]
```

```
The array is no longer needed.
Deleting the array and releasing its memory...
Memory successfully deleted and released.
```

```
=====
Would you like to create another array?
```

```
[Y] Yes, create another array
```

```
[N] No, end the program
```

```
Enter your choice: n
```

```
=====
Thank you for using our dynamic array program.
COMP 371 Assignment 1 – Part 1 completed successfully.
=====
```

Part 2

```
wissemoumsalem@Wissems-MacBook-Air comp371-A1 % g++ -std=c++11 part2.cpp -o part2
./part2
```

COMP 371 - Assignment 1, Part 2
Reema Aboudraz and Wissem Oumsalem

1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit

Choice: 1

Enter point 1:

x: 0
y: 0
z: 0

Enter point 2:

x: 3
y: 0
z: 0

Enter point 3:

x: 0
y: 4
z: 0

Triangle created.

1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit

Choice: 2

Vertex 1: (0, 0, 0)
Vertex 2: (3, 0, 0)
Vertex 3: (0, 4, 0)

1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit

Choice: 4
Area: 6

1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit

Choice: 3
Distance: 2
Axis (x, y or z): y
Triangle translated.

```
1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit
Choice: 2
Vertex 1: (0, 2, 0)
Vertex 2: (3, 2, 0)
Vertex 3: (0, 6, 0)
```

```
1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit
Choice: 3
Distance: 5
Axis (x, y or z): w
Invalid axis.
```

```
1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit
Choice: 9
Invalid choice.
```

```
1. Create a triangle
2. Display the triangle
3. Translate the triangle
4. Calculate the area
5. Exit
Choice: 5
Program ended.
```

```
○ wissemoumsalem@Wissems-MacBook-Air comp371-A1 %
```